MILITARY SPECIFICATION

PANEL ASSEMBLIES: 10548439 AND 10549788

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers two types of transistorized panel assemblies. They provide voltage control, distribution, regulating, and battery charging functions when energized with a direct current of between 18 volts and 30 volts. The panel assemblies also provide voltage control, regulating, and distribution functions when energized from an emergency direct current six volt battery power source.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on data of invitation for-bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacturing and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

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DRAWINGS

U.S. Army, Frankford Arsenal

F10548439	Panel Assembly
F10549788	Panel Assembly

PACKAGING DATA SHEETS

PS10548439	Packaging of Panel Assembly: 10548439
PS10549788	Packaging of Panel Assembly: 10549788

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with the drawings, specifications, and standards which form a part of this specification.

3.2 Fabrication. - The panel assembly shall be manufactured in accordance with the applicable drawing listed in Section 2 and drawings pertaining thereto.

3.3 General specification. - The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.4 Environmental service conditions. - Unless otherwise specified, the panel assembly shall meet all requirements of this specification at an ambient temperature between +60° and +90° Fahrenheit (F).

3.4.1 Vibration. - The panel assembly shall show no evidence of physical failure due to being vibrated successively along three (3) mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either linearly or logarithmically from 5 to 55 to 5 CPS in five (5) minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64 inch (1/32 inch total excursion) (See note under Table IV).

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3.4.2 Shock. - The panel assembly shall show no evidence of physical failure after being subjected to a minimum of 6 shocks of at least 30 gravitational units (g's) with each shock being 11 ± 1 milliseconds duration.

3.4.3 Operating temperatures. - The panel assembly shall meet all the requirements of 3.6 while thermally stabilized at temperatures of -65° and +150°F.

3.4.4 Storage temperatures.- After having been thermally stabilized at temperatures of -80° and +160°F, the panel assembly shall show no evidence of damage and shall be capable of meeting all requirements of this specification at ambient temperatures between 60° and 90°F.

3.5 Electrical service conditions.

3.5.1 Primary power source. - The panel assembly shall meet the applicable requirements of 3.6 when energized with a primary power source of 18 VDC to 30 VDC. This voltage shall be applied in positive polarity to J1-H with respect to J1-K in accordance with Drawing F10542023 for Panel Assembly 10548439, and Panel Assembly 10549788.

3.5.2 Emergency power source. - The requirements of 3.6, applicable to this source, shall be met when the panel assembly is energized with an emergency power source of 6.0 VDC. This voltage shall be applied in positive polarity to J1-E with respect to J1-D, for the applicable panel assembly.

3.5.3 Input immunity to reverse polarity. - The panel assembly Shall not be damaged by the application of electrical potentials of 3.5.1 in reverse polarity.

3.5.4 Ground potential. - When energized as specified in 3.5.1 exposed metallic surfaces of the panel assembly enclosure, exclusive of electrical terminations and sub-surface mounted parts, shall be at ground potential.

3.6 Performance.

3.6.1 Rheostat performance. - The panel lamps (R1), reticle projector (R2), and wiper (R3) rheostats when rotated from an extreme clockwise (cw) to extreme counterclockwise (ccw) position shall exhibit no spasmodic performance throughout their range of rotation.

3.6.1.1 Rheostat R1. - Rheostat R1 shall control total current as specified in Table 1, for each respective rheostat arm position and applied primary power source voltage. Switch S1 shall be in the "OFF" position when meeting this requirement. Primary power shall be applied as follows: +30 VDC to J1-H, GRD to J1-J.

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TABLE I - R1 PERFORMANCE

Rheostat	Primary Power	Rheostat Wiper arm position	Circuit Load DC amps
R1	30 VDC	extreme CW - Min. Res.	.28 ± .04
R1	30 VDc	extreme CCW - Min. Res.	.185 ± .01
R1	30 VDc	off	.12 ± .01

3.6.1.2 Rheostat R2. - Rheostat R2 output circuit shall exhibit the voltage specified in Table II for each respective circuit under the specified DC load conditions, rheostat arm positions, and applied primary power, Switch S1 shall be in the "Emergency" position.

TABLE II - R2 PERFORMANCE

Rheostat	Primary Power	Pheostat Wiper Arm Position	Circuit Load DC AmpS	Circuit Out puts DC volts
R2	6 VDC	extreme CW - Min. Res.	.700	5.6 ± .4
R2	6 VDC	extreme CCW - Max. Res.	---	1.7 ± .2
R2	6 VDC	extreme OFF position	---	Zero

3.6.1.3 Rheostat R3. - When the panel assembly is energized from a primary power source of +30 VDC, the output circuits of R3 (existing between chassis ground and terminal J1-A and J1-B) shall exhibit the voltages specified in Table III at the specified circuit load conditions and rheostat wiper arm positions.

TABLE III - R3 PERFORMANCE

Rheostat Wiper Arm Position	Circuit Load DC Amps	Output Circuit Voltage J1-A to Grd.	J1-B to Grd.
extreme W-Min. Res.	1.00	29.0 VDC Min.	29.0 VDC Min.
CCW-to Max. Res.	.60 max.	29.0 VDC Min.	15.0 VDC Min.
* extreme CCW-OFF Position	Zero	29.0 VDC Min.	Zero M1.n.

* Switch S1 shall be in the "center-off" position.

3.6.1.4 Output of voltage regulated supplies. - The battery charging, reticle projector, and high voltage oscillator supplies shall exhibit the voltage regulation specified in Table IV. This regulation shall exist over the indicated primary power source voltage and under the specified DC load conditions. The reticle projector and high-voltage oscillator outputs in Table IV shall only be available with Switch S1 in the "ON" position.

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With S1 in the "OFF" position Terminals J1-K and J1-L shall exhibit zero voltage relative to all ground terminals.

TABLE IV - REGULATED OUTPUTS

Regulated Supply	S1 Position	Primary Power	Output Load	Required Output DC
Battery Charging	Off (Center)	30 VDC	732 OHMS ± 1%	7.0 ± 0.5 Volts
Battery Charging	Off (Center)	18 VDC	Zero OHMS (Short Circuit)	220-320 M.A.
Battery Charging	On (Veh. Pwr.)	18 VDC	Zero OHMS (Short Circuit)	220-320 M.A.
Reticle Projector	On (Veh. Pwr.)	30VDc/18VDC	732 OHMS ± 1%	7.3 ± 0.5 Volts
Reticle Projector	On (Veh. Pwr.)	18VDC/30VDC	800 M.A.	Less than .30 Volt Change
Hight Volt. Oscill.	On (Veh. Pwr.)	18-30 VDC	10-20 M.A.	5.9 ± 0.1 Volts
Reticle Projector	On (Emer. Pwr.)	6.2 VDC	0-600 M.A.	5.8 ± 0.4 Volts
High Volt. Oscill.	On (Emer. Pwr.)	6.2 VDC	10-20 M.A.	5.9 ± 0.1 volts

These requirements shall be met with R2 in the clockwise position (Min. resistance).
These requirements shall also be met during vibration requirements of 3.4.1.

3.6.1.5 Output voltage rise time. - Panel assembly output voltages shall be available within one second after application of the required energizing potential.

3.6.1.6 Recovery time. - Panel assembly regulated output voltage levels shall exhibit no time delay in maintaining a constant output voltage level with sudden changes in DC output loads.

3.6.2 Battery charging circuit. - The battery charging circuit, operating on a primary power source input voltage of 18 to 30 VDC, shall be capable of supplying a charging current up to .320 milliamps to the battery within one second after application of the input voltage.

3.6.3 Controls. - Controls shall operate freely and smoothly without binding under all operational service conditions. Switch type controls shall indicate positive indexing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified. the supplier may utilize his own facilities or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 panel assemblies plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIIATARY-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables V, VI, and VII herein.

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Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables V, VI, and VII shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE V - CLASSIFICATION OF DEFECTS

CLASS
CRITICAL: NONE
MAJOR: AQL 1.0% Defective

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Vibration	3.4.1	4.6.4
102.	Input immunity to reverse polarity	3.5.3	4.6.5.1
103.	Ground potential	3.5.4	4.6.5.2
104.	Rheostat performance	3.6.1	4.6.5.3
105.	Rheostat R1	3.6.1.1	4.6.5.4
106.	Rheostat R2	3.6.1.2	4.6.5.5
107.	Rheostat R3	3.6.1.3	4.6.5.6
108.	Regulated output, switch S1 "OFF"	3.6.1.4	4.6.5.7.1
109.	Regulated output, 30VDC with minimum load	3.6.1.4	4.6.5.7.2
110.	Regulated output, 18 VDC with minimum load	3.6.1.4	4.6.5.7.3
111.	Regulated output, 18 VDC with maximum load	3.6.1.4	4.6.5.7.4
112.	Regulated output, 30 VDC with maximum load	3.6.1.4	4.6.5.7.5
113.	Regulated output, 6.2 VDC with maximum load	3.6.1.4	4.6.5.7.6
114.	Regulated output, 6.2 VDC with minimum load	3.6.1.4	4.6.5.7.7
115.	Output voltage rise time	3.6.1.5	4.6.5.8
116.	Recovery time	3.6.1.6	4.6.5.9
117.	Battery charging circuit	3.6.2	4.6.5.10
118.	Controls	3.6.3	4.6.5.11

MINOR: NONE

NOTE: The tests in Table V shall be conducted at a tempeature between +60° and +90°F.

4.3.3 Acceptance and reflection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejections" specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General. - One panel assembly shall be selected at random as a special sample from each 100 produced, or from each month's production whichever occurs firsts The sample shall meet the requirements and tests in Table VI.

TABLE VI

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable drawing - visual
302.	General specification	3.3	MIL=F-13926 - visual

4.4.2 Environmental. - Three panel assemblies shall be selected at random as samples from each 60 produced, or from each month's productions whichever occurs first. Each sample shall meet the requirements and tests in Table VII and shall then meet the requirements and tests in Table V.

TABLE VII

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° to +160°)	3.4.4	4.6.2
304.	Rheostat performance	3.4.3, 3.6.1	4.6.3, 4.6.5.3
305.	Rheostat Rl	3.4.3, 3.5.1.1	4.6.3, 4.6.5.4
306.	Rheostat R2	3.4.3, 3.6.1.2	4.6.3, 4.6.5.5
307.	Rheostat R3	3.4.3, 3.6.1.3	4.6.3, 4.6.5.6
308.	Regulated output, switch S1 "OFF"	3.4.3, 3.6.1.4	4.6.3, 4.6.5.7.1
309.	Regulated output, 30 VDC with minimum load	3.4.3, 3.6.1.4	4.6.3, 4.6.5.7.2
310 .	Regulated output, 18 VDC with minimum load	3.4.3, 3.6.1.4	4.6.3, 4.6.5.7.3
311.	Regulated output, 18 VDC with maximum load	3.4.3, 3.6.1.4	4.6.3, 4.6.5.7.4
312.	Regulated output, 30 VDC with maximum load	3.4.3, 3.6.1.4	4.6.3, 4.6.5.7.5
313.	Regulated output, 6.2 VDC with minimum load	3.4.3, 3.6.1.4	4.6.3, 4.6.5.7.6
314.	Regulated output, 6.2 VDC with minimum load	3.4.3, 3.6.1.4	4.6.3, 4.6.5.7.9
315.	Output voltage rise time	3.4.3, 3.6.1.5	4.6.3, 4.6.5.8
316.	Recovery time	3.4.3, 3.6.1.6	4.6.3, 4.6.5.9
317.	Battery charging circuit	3.4.3, 3.6.2	4.6.3, 4.6.5.10
318.	Controls	3.4.3, 3.6.3	4.6.3, 4.6.5.11
319.	Shock	3.4.2	4.6.1

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NOTE: The tests for characteristics 304 through 318 inclusive, in Table VII, shall be conducted at temperatures of -65° and +150°. The test for characteristic 319 shall be conducted at a temperature between +60° and +90°F.

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-1-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-1-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Vibrating machine. - The vibrating machine shall be capable of providing the amplitude and frequency specified in 3.4.1. The vibrating machine shall also be capable of vibrating the panel assembly along 3 mutually perpendicular axes while the panel assembly is maintained with its longitudinal axis in a horizontal plan.

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4.5.2.3.2 Primary power source. - When a primary power input of 18 VDC to 30 VDC is required for tests, the regulated power supply specified in 4.5.3 shall be connected to the panel assembly as indicated in 3.5.1. The automatic DC digital voltmeter shall be utilized for the verification of the output voltage when the power supply is adjusted from 18 VDC to 30 VDC.

4.5.2.3.3 Emergency primary power source. - When a primary power input of 6.0 VDC is required for tests, the regulated power supply specified in 4.5.3 shall be connected to the panel assembly as indicated in 3.5.2. The automatic DC digital voltmeter shall be utilized for the verification of the output voltage.

4.5.3 Test equipment. - In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

NOMENCLATURE	DESCRIPTION
Regulated voltage supply	Wide input voltage and frequency range: 105 - 135 vac, 45 - 480 cycles per second. Regulation: .015% or 1 millivolt (line or load) Ripple: 250 microvolt RMS, 1 millivolt peak-to-peak. Voltage range continuously variable between 0-40 VDC. Current: 4.0 amperes.
DC milliammeter, clip-on	Accuracy: ± 2% of full scale
Automatic DC digital voltmeter	Accuracy: ± 2% of reading
Oscilloscope	Accuracy: ±1% of reading {peak-to-peak)

4.6 Test methods and procedures.

4.6.1 Shock. - The panel assembly shall be shocked in accordance with Procedure II of MIL-F-13926 except, the number of impact shocks and minimum shock pulse (g's) shall be as specified in 3.4.2. The shock pulses (a series of 6 total) shall be applied, 1 shock to each of the 6 surfaces of the panel assembly. At the conclusion of the shock test the panel assembly shall be subjected to the visual, tactile, and audible examination of 4.6.4.1. The panel assembly shall conform to the

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requirements of 3.4.2. Upon successful completion of this examination, the panel assembly shall be subjected to the tests specified in 4.6.2 through 4.6.5.11 inclusive.

4.6.2 Storage temperatures. - Testing equipment utilized in this test shall be in accordance with the "Test Facilities"' requirements of MIL-F-13926 and the conditions of 3.4.4. The panel assembly shall be exposed to each specified temperature for a minimum of 5 hours. Subsequent to the temperature exposures, the panel assembly shall be stabilized in an ambient temperature between +60° and +90° Fahrenheit and subjected to the visual, tactile, and audible examination of 4.6.4.1. The panel assembly shall show no evidence of damage. Upon successful completion of this examination the panel assembly shall be subjected to the tests specified in 4.6.3 through 4.6.5.11 to determine overall conformance with 3.4.4.

4.6.3 Operating temperatures. - This test may be performed in conjunction with the test in 4.6.2 utilizing the testing equipment specified therein. The conditions of 3.4.3 shall apply. The panel assembly shall be exposed to each temperature extreme specified in 3.4.3 for a minimum of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the panel assembly shall be subjected to the tests specified in 4.6.5 to 4.6.5.11 inclusive. The panel assembly shall conform to the requirements of 3.4.3.

4.6.4 Vibration. - Perform this test with the panel assembly positioned on special testing equipment that conforms to 4.5.2.3.1. Energize the panel assembly in accordance with the procedure defined in 4.6.5.7. The panel assembly shall be vibrated in accordance with Procedure IX, Part 1 of MIL-STD-810, except that the frequency, frequency cycling and amplitude of vibration shall be as specified in 3.4.1. During the vibration cycle, the high voltage oscillator circuit of the panel assembly shall be subjected to the tests specified in 4.6.5.7.1 through 4.6.5.7.7 inclusive, and shall conform to the applicable requirements specified therein (see note under Table IV). At the conclusion of the vibration test de-energize the panel assembly and subject it to the test in 4.6.4.1.

4.6.4.1 Examination. - Subsequent to vibration, the panel assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials, to determine compliance with 3.4.1, prior to being subjected to the remaining tests specified herein. A visual and tactile examination shall be made for missing or damaged parts and to check functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall be made for loose or broken internal parts and solder. Any one of the possible defects defined above shall be cause to consider a panel assembly defective. Upon successful completion of this examination the panel assembly shall then be subjected to the test specified in 4.6.5 to 4.6.5.11 inclusive to determine overall compliance with 3.4.1.

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4.6.5 Performance. - The following tests shall be performed with the regulated voltage supply defined under 4.5.3 connected to the panel assembly as indicated in 3.5.1 and 3.5.2. Unless otherwise specified, the required power source voltages shall be applied in positive polarity. The remaining test equipment defined under 4.5.3 shall be utilized to measure the applicable output performance of the panel assembly.

4.6.5.1 Input immunity to reverse polarity. - Conduct this test in sequence to, and upon successful completion of the tests specified in 4.6.5.7.1 to 4.6.5.7.5 inclusive. Perform the test for input immunity by reversing the primary power source polarity to negative and energize the panel assembly with the required potential for a minimum of 30 seconds. At the end of the 30 second period, return the primary power source polarity to positive. Repeat the tests of 4.6.5.7.1 to 4.6.5.7.7 inclusive and note any difference in the regulated output voltages readings from the previous measurements. The panel assembly shall show no evidence of damage as indicated by the regulated output voltage measurements, to determine compliance with the requirements of 3.5.3. This test shall be performed at the temperature specified in 3.4.

4.6.5.2 Ground potential. - This test shall be performed utilizing the digital voltmeter specified in 4.5.3 to verify the ground potential of the exposed metallic surfaces of the panel assembly enclosure, exclusive of electrical terminations and sub-surface mounted parts. The panel assembly shall conform to the requirements of 3.5.4. This test shall be performed at the temperature specified in 3.4.

4.6.5.3 Rheostat performance. - Adjust the regulated voltage supply to apply VDC to each of the specified rheostat circuits as designated in 3.6.1. For each respective circuit load, the respective rheostat knob shall be rotated as specified in 3.6.1. Throughout the range of rotation, for each rheostat knob, observe the digital voltmeter to determine compliance with the requirements of 3.6.1, at the temperatures indicated in 3.4 and 3.4.3.

4.6.5.4 Rheostat R1. - This test shall be performed with the primary power source voltages in Table I applied as specified in 3.6.1.1. Switch S1 shall be in the “OFF” position during this test. The total measured current controlled by Rheostat R1 for each respective rheostat wiper arm position and applied primary power source voltage shall be within the tolerance specified in Table I of 3.6.1.1, at the temperatures indicated in 3.4 and 3.4.3.

4.6.5.5 Rheostat R2. - This test shall be performed with the primary power source voltages in Table II applied as specified in 3.6.1.2. Switch S1 shall be in the “EMERGENCY POWER ON” position during this test. The measured Rheostat R2 output voltage for each respective circuit under the specified DC load conditions, rheostat wiper arm positions, and applied primary power shall exhibit the voltages specified in Table II of 3.6.1.2, at the temperatures indicated in 3.4 and 3.4.3.

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4.6.5.6 Rheostat R3. - This test shall be performed with the panel assembly energized as specified in 3.6.1.36. Switch S1 shall be in the "CENTER OFF" position during this test. The measured R3 output voltage for each respective circuit under the specified DC load conditions, rheostat wiper arm positions, and applied primary power source shall exhibit the voltages specified in Table III of 3.6.1.3, at the temperatures indicated in 3.4 and 3.4.3.

4.6.5.7 Output of voltage regulated supplies. - Prior to conducting the tests of the regulated outputs in Table IV the following preliminary adjustments shall be made to the panel assembly:

a. Set the input voltage to +24 VDC at J1-H.
b. Set the battery charging circuit to maximum load (shorted battery) condition.
c. Set the reticle projector load to 800 ma.
d. Set the high voltage oscillator load to 20 ma.
e. Adjust the trimpot on TB2 to provide +5.9 volts output at J1-L.

Subsequent to the preliminary adjustments, the panel assembly shall be energized and subjected to the following tests. The applicable primary power input voltages required in Table IV shall be applied as specified in 3.5.1 and 3.5.2. The applicable circuits shall be tested at minimum and maximum voltage inputs, and minimum and maximum loads as indicated in Table IV, and the following testing notes.

TESTING NOTES AND DEFINITIONS

Battery charging circuit = With battery fully charged battery load current = 0 amps.

With battery fully discharged battery load current = .320 amps. max.

Testing under vehicle power (18-30VDC)

Total max. load = Reticle (.800 amps) + secondary regulator (.020 amps) + Battery (.320 amps) = 1.140 amps

Min. load = No current drawn by reticle or battery secondary regulator will draw .010 amp

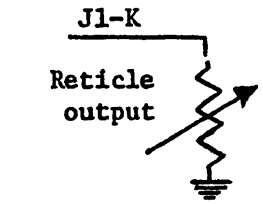
Testing under emergency power (6.2 VDC).

Max. load = Reticle (.600 amps) + secondary regulator (.020 SMPS) = .620 ampS

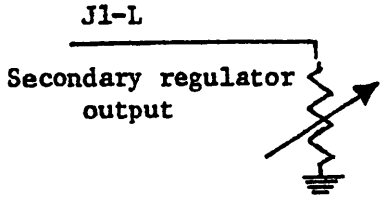
Min. load = Secondary regulator (.010) - .010 amps

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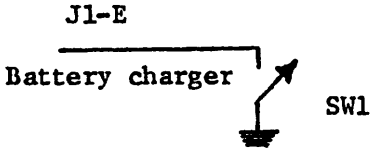
Test setup-outputs



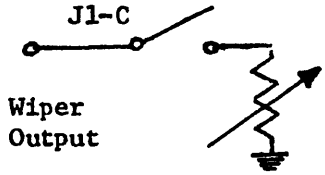
Adjust to obtain max. & min. load as specified



Adjust to obtain max. & min. load as specified



SW on = max. load
SW off = min. load



Adjust to obtain max. and min. load as specified.

N0TE: Use leads as short as possible when testing under emergency power conditions.

4.6.5.7.1 Regulated output Switch S1 OFF. - With Switch S1 in the OFF position, load the battery charging circuit to the minimum and maximum resistance values in Table V and measure for the applicable voltage and current values from Table IV. Test for the disconnection of the oscillator and reticle voltage outputs as designated in 3.6.1.4.

4.6.5.7.2 Regulated output, 30 VDC input with minimum loads. - Apply 30 VDC to the panel assembly with S1 in the "Vehicle Power ON" position with the battery charging, high voltage oscillator and reticle projector outputs loaded to the minimum value. Rotate Rheostat R2 to the extreme clockwise position. Measure and record the voltage outputs of the reticle projector and high voltage oscillator circuits. The regulated output voltages of these circuits shall conform to the values of Table IV of 3.6.1.4.

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4.6.5.7.3 Regulated output, 18 VDC input with minimum loads. - Energize the panel with 18 VDC and perform the test as specified in 4.8.7.3.5.2. The output voltage measured shall conform to the values of Table IV of 3.6.1.4.

4.6.5.7.4 Regulated output 18 VDC input with maximum loads. - With the panel energized as specified in 4.8.7.3.5.3, increase the load on the reticle projector output to maximum. Measure the voltage output of the reticle projector supply at Pin K. The change of output voltage from the measurement of 4.6.5.7.3 shall be equal to or less than the maximum change specified in Table IV of 3.6.1.4. Increase the load on the battery charging circuit to maximum. The measured output current value shall conform to Table IV of 3.6.1.4. Increase the load on the high voltage oscillator circuit to maximum. The measured voltage output shall conform to the value in Table IV of 3.6.1.4. Return the loads on the reticle projector, high voltage oscillator and battery charging circuit to minimum value.

4.6.5.7.5 Regulated output 30 VDC input with maximum loads. - Energize the panel with 30 VDC. Increase the load on the reticle projector output to maximum. Measure the voltage output of the reticle projector supply at Pin K. The change of output voltage from the measurement of 4.6.5.7.2 shall be equal to or less-than the maximum change specified in Table IV of 3.6.1.4. Increase the load on the high voltage oscillator circuit. The measured voltage output shall conform to the value in Table IV of 3.6.1.4.

4.6.5.7.6 Regulated output, 6.2 VDC with maximum load. - perform this test with Switch S1 in the "EMERGENCY POWER-ON" position and +6.2 VDC applied from J1-E to ground (+6.2 VDC at J1-11, -6.2 VDC at ground); the Rheostat R2 in the extreme clockwise position and the reticle projector circuit loaded to maximum load (.600 amp); the high voltage oscillator circuit loaded to maximum load (.020 amp). Measure the output voltage of the reticle projector and high voltage oscillator circuits. The regulated output voltage of these circuits shall conform to the applicable voltage values in Table IV of 3.6.1.4, at the temperatures indicated in 3.4 and 3.4.3.

4.6.5.7.7 Regulated output, 6.2 VDC with minimum load. - Perform this test with the Switch S1 in the "EMERGENCY POWER-ON" position, and the 6.2 VDC applied as in 4.6.5.7.6; the Rheostat R2 in the extreme clockwise position and the reticle projector circuit loaded to minimum load (open circuit); the high voltage oscillator circuit loaded to minimum load (.010 amp). Measure the output voltage of the reticle projector and high voltage oscillator circuits. The regulated output voltage of these circuits shall conform to the applicable voltage values in Table IV of 3.6.1.4, at the temperatures indicated in 3.4 and 3.4.3.

4.6.5.8 Output voltage rise time.- This test shall be performed in conjunction with tests of 4.6.5.7.1 to 4.6.5.7.7 inclusive. Output voltage rise time for each regulated supply shall be available within one second after application of the required energizing potential, as measured on the

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oscilloscope specified in 4.5.3 to determine compliance with 3.6.1.5. The output voltage rise time shall meet the requirements of 3.6.1.5 at the temperatures indicated in 3.4 end 3.4.3.

4.6.5.9 Recovery time. - This test shall be performed in sequence to 4.6.5.8. While each regulated supply is energized with its required potential, Rheostats R1, R2 and R3 shall be rotated from an extreme clockwise position. During the range of rotation of each rheostat wiper arm, the panel assembly regulated output voltage levels shall conform to the requirements of 3.6.1.6, as measured with the oscilloscope. The recovery time shall meet the requirements of 3.6.1.6 at the temperatures indicated in 3.4 and 3.4.3.

4.6.5.10 Battery charging circuit. - This test shall be performed in conjunction with 4.6.5.7. The battery charging circuit supply shall be tested utilizing the oscilloscope and millimeter to measure the time and charging current. The circuit supply shall be tested at minimum and maximum input voltage to the panel assembly. The battery charging circuit shall conform to the requirements of 3.6.2. The battery charging circuit shall meet the requirements of 3.6.2 at the temperatures indicated in 3.4 and 3.4.3.

4.6.5.11 Controls. - The controls shall be subjected to a visual and tactile examination to determine compliance with 3.6.3. The panel assembly controls shall meet the requirements of 3.6.3 at the temperature indicated in 3.4 and 3.4.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with the applicable Packaging Data Sheet listed in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The panels are primarily intended for use with Tank Periscope: XM44E1 or X1144E2. They are intended to supply a regulated DC output voltage to energize the high voltage oscillator of these periscopes and to provide a charging current for the stand-by battery employed. The panels are also intended to provide distribution of an adjustable voltage for panel assembly and reticle projector illumination and wiper motor operation.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and data of this specification.

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b. Applicable stock number.
c. Selection of applicable levels of preservation, packaging and packing.
d. Requirements for First Article submission.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A575

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-P-60243C(MU) Panel Assemblies: 10548439 and 10549788

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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